

Marketing

Chapter 4: Price

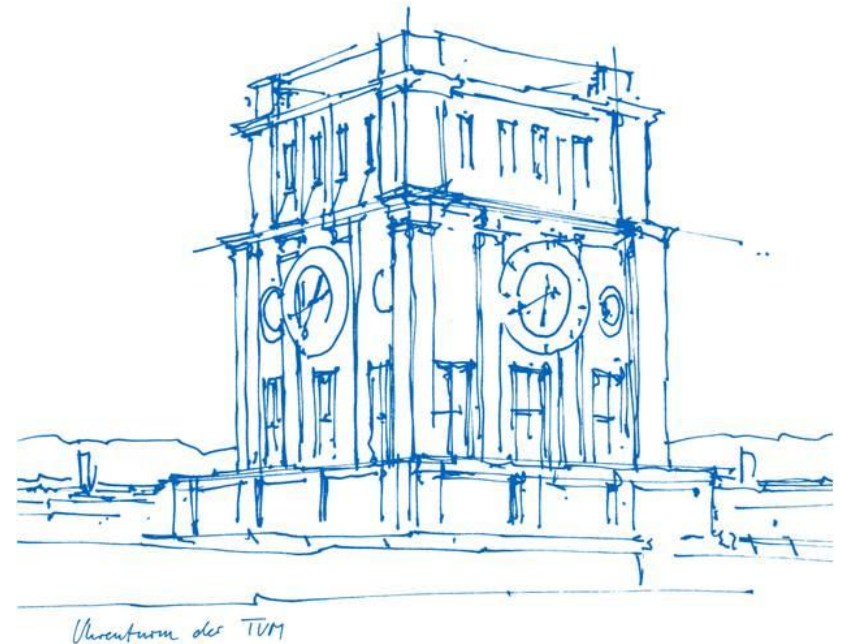
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Technical University of Munich

TUM School of Management

Center for Digital Transformation

Professorship of Digital Marketing



Learning goals

After studying this chapter, you will be able to:

- Explain the central characteristics and elements of pricing strategies
- Name pricing instruments
- Differentiate strategic pricing decisions
- Explain price differentiation and its forms
- Explain price-bundling strategy as a special type of price differentiation
- Understand and distinguish methods of price determination (cost-oriented, marginal analytical)

Price is a powerful sword

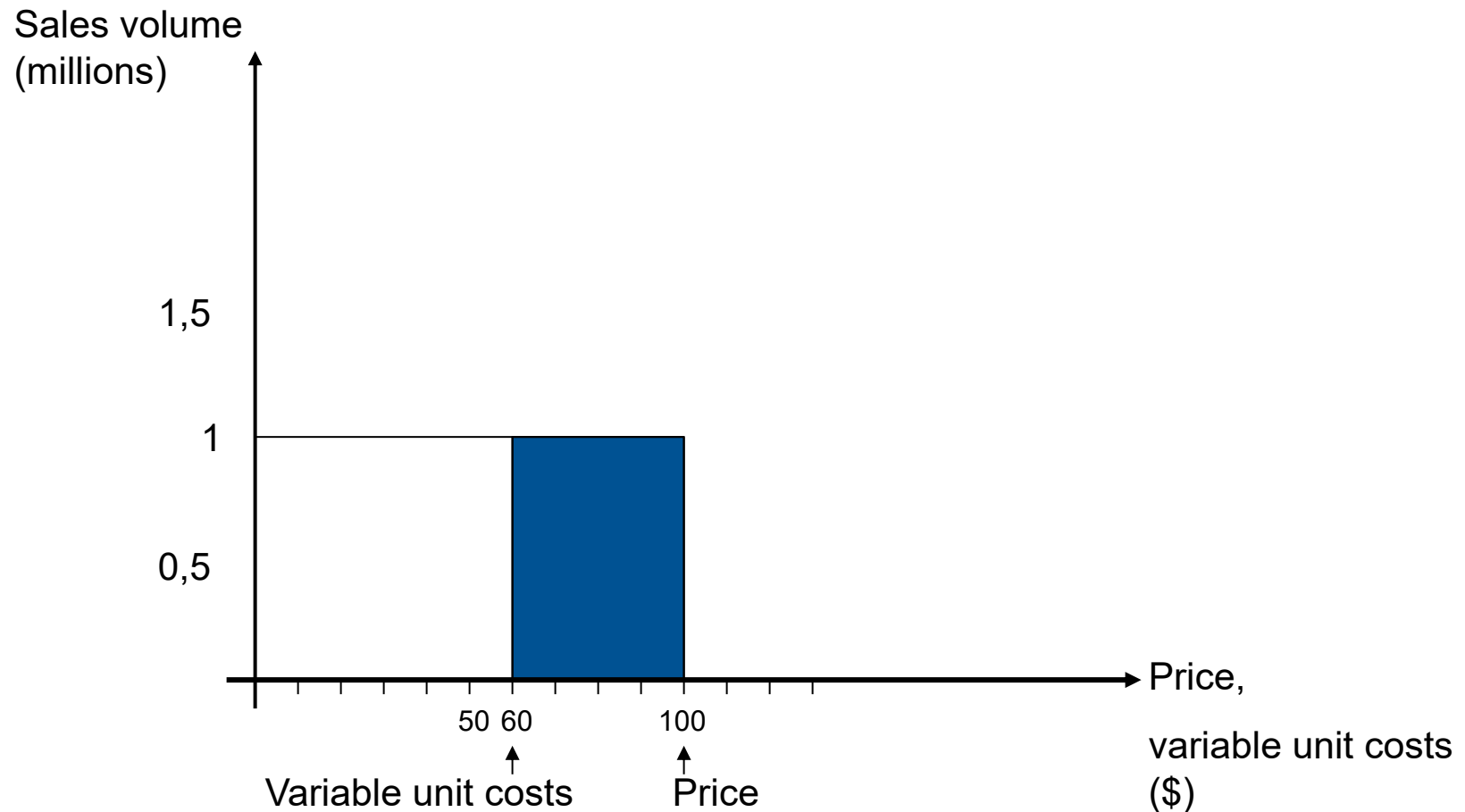
A 10% increase/decrease in...



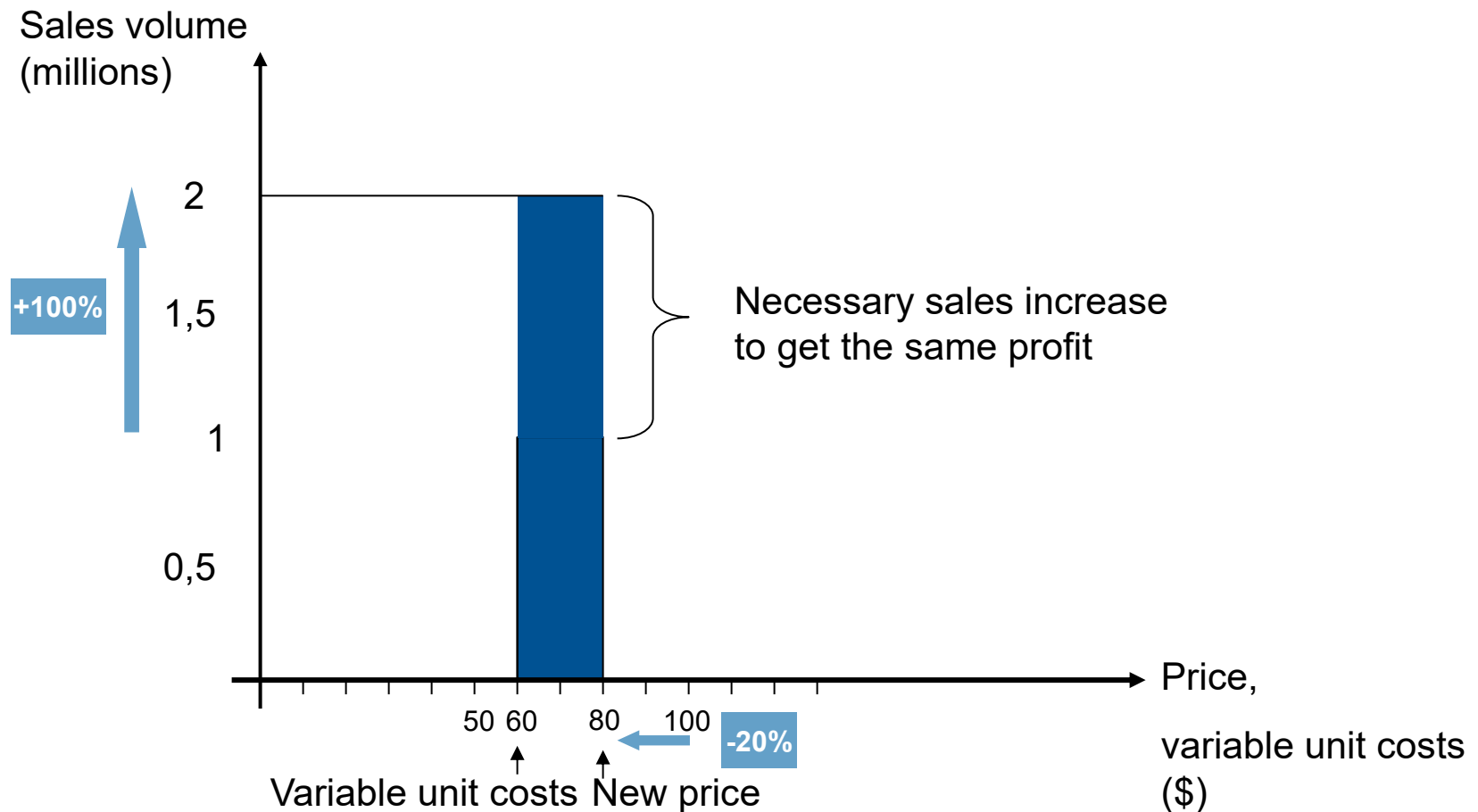
			Profit (in \$m)		
	Old	New	Old	New	
Price (in €)	100	110	10	20	+ 100%
Variable costs (in €)	60	54	10	16	+ 60%
Sales	1m	1.1m	10	14	+ 40%
Fixed costs (in €)	30m	27m	10	13	+ 30%

Price is a powerful sword

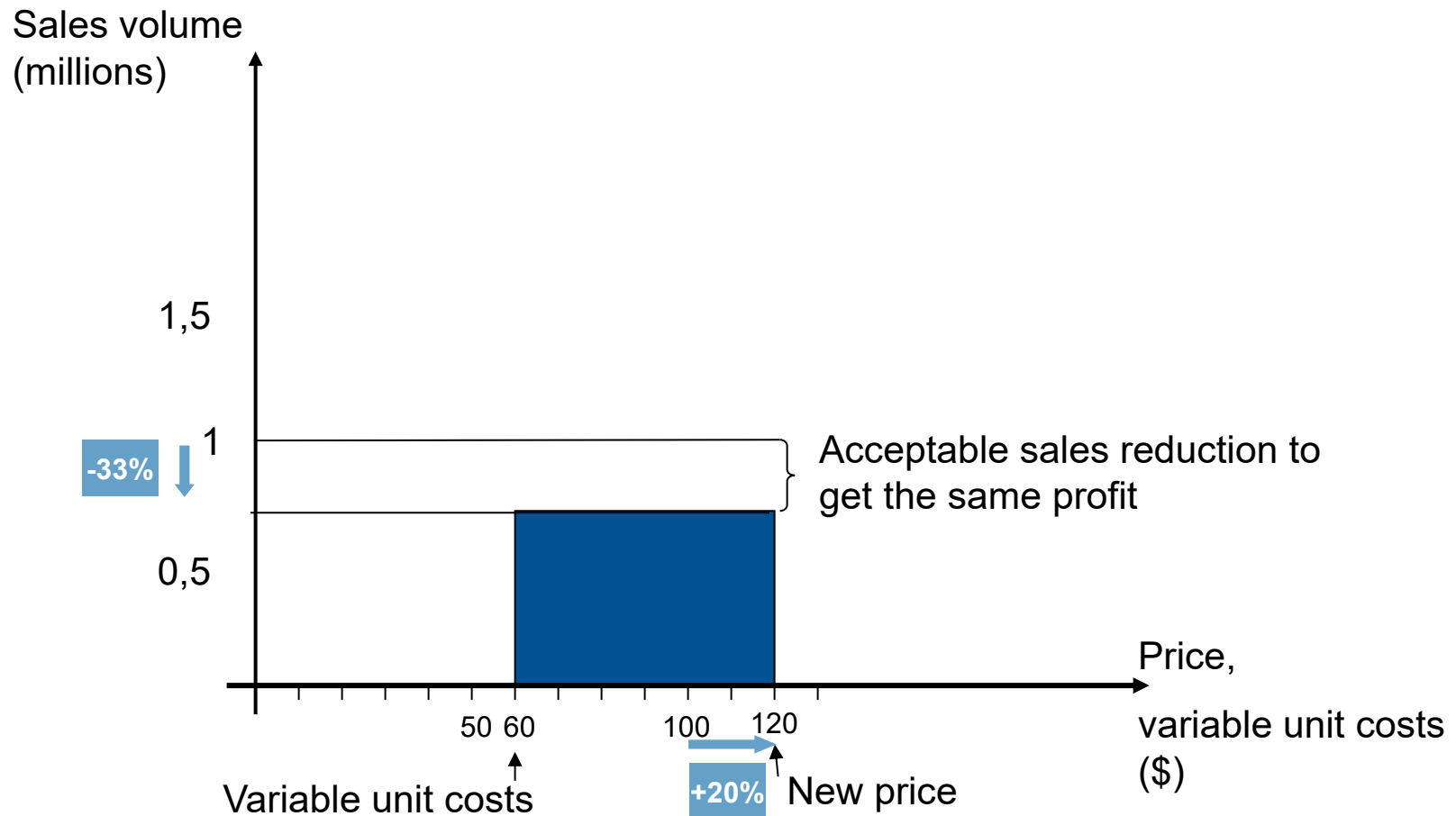
Initial situation



Price is a powerful sword



Price is a powerful sword



Behavioral pricing

Which options would you buy?



0,3 liter



0,5 liter

Behavioral pricing

Which options would you buy?



0,3 liter



0,5 liter



0,75 liter

Compromise effect

"The highest margins come from drinks and chips. That's why McDonalds wants customers to order menus wherever possible. Originally, there were only small cups with 0.3 liters and large ones with 0.5 liters. **Six out of ten customers opted for the small drink at the time.**

So, we introduced 0.75-liter cups and asked the customers whether they wanted a small, medium or large drink. **Suddenly most of them took 0.5 liters.**

Not because they really wanted to drink so much, but because customers usually choose the medium portion size."



Compromise effect – Other examples



50%

50%

Compromise effect – Other examples



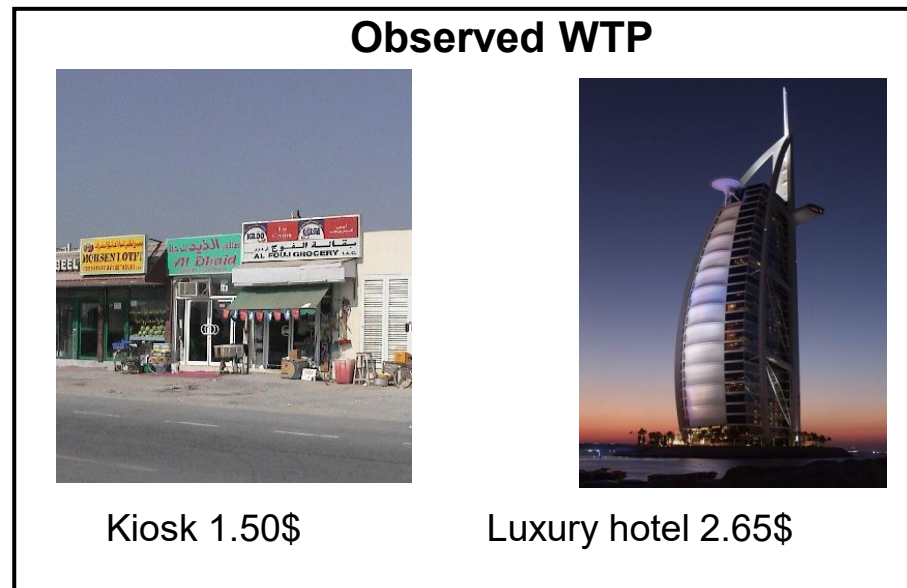
30%

60%

10%

Behavioral pricing – key insights for marketers

Consumers' WTP is influenced by **price expectations** and perceived **price fairness**.



Thaler (1985), pp. 206

Price expectations:

customers expect different prices at a local store vs. a luxury hotel.

Price fairness:

Customers are aware of higher fixed costs in hotels (e.g., higher personnel costs) that "justify" higher prices.

Behavioral pricing – "Pain of Paying: The psychology of money"



Youtube video: <https://www.youtube.com/watch?v=PCujWv7Mc8o>

Behavioral pricing

Would you be willing to drive to another store to save 10€ on a Bluetooth speaker?

Media-Markt



29€



Travel time:
10 minutes

Saturn



19€

Behavioral pricing

Would you be willing to drive to another store to save 10€ on a new TV?

Media-Markt



1299€



Travel time:
10 minutes

Saturn



1289€

Behavioral pricing – Key insights for marketers

The perceived extent of price changes (e.g., price reductions) is evaluated based on **relative** rather than **absolute** price changes.

Based on standard economic theories, the utility (perceived value) of money is constant.

→ 10€ should be valued equally in all transactions.

Studies show that 34% customers would be willing to travel 10 minutes to save 10€ on the bluetooth speaker (phone), while less than 1% of the respondents would travel 10 minutes to save 10€ on the TV set.

→ Concepts in behavioral pricing help one to understand real-world decision-making.

Behavioral pricing

Decoy effects

Which one do you prefer?



Which one do you prefer now?



Price information acquisition, evaluation, and storing



Price information acquisition is a process that leads to price knowledge that is stored in the short-term memory.

Price information evaluation is a price assessment process in short-term memory that includes the decoding, mental processing, and evaluation of the perceived price information.

Price information storing describes learning processes; price information is stored in long-term memory ("prices are learnt") and can be retrieved by the short-term memory.

Price information acquisition is based on price interest

Consumer **intensify** information search if...

1. Value for money risk is high (e.g., new technology – Samsung Galaxy Fold)
2. Information search is simple and easy (e.g., on price comparison websites)
3. Product price is relatively high (e.g., expensive luxury car)

Price information acquisition behavior

- Consumers often do not search for price information upfront but at the point-of-sale (PoS; PoP); **little between-store-search**
- Even within a store, there is often **no active price search**; product evaluation is based on price information that is easy to find (passive information acquisition)
- Demand is aligned based on price information; people buy what they **perceive** to be reasonably priced
- People rely on **generalizable decision heuristics** (e.g., compromise effect, package size, brands as quality signals that justify a higher price, etc.)

Meffert, Burmann, Kirchgeorg, Eisenbeiß (2019)

Price information acquisition – Marketers' strategies

1. **Cost leader:** communication strategy highlights value-for-money advantage
(increase customers' price interest)
2. **Value strategy:** focus on communicating quality and service
(decrease customers' price interest by highlighting other benefits)
3. Try to **guide customers' price interest** by introducing a cheaper secondary brand



Meffert, Burmann, Kirchgeorg, Eisenbeiß (2019)

Price information acquisition – Marketers' strategies

3. Try to **guide customers' price interest** by introducing a cheaper secondary brand

Apple Inc. released five new iPhone models in 2020, the most in the device's history. Apple started in 2007 with one iPhone but soon **established a "good-better-best" strategy**, where budget and premium products flank "Goldilocks" options.

For years, the company did this by keeping older models around at slashed prices. In recent years, it has introduced more new options at various price tiers. This year's lineup ranged from the \$399 late-adopter-targeted **iPhone SE** to the feature-packed \$1,099 **iPhone 12 Pro Max**—with a "just right" iPhone 12 in the middle. The company also gave Apple Watch shoppers three tiers of options for the first time.

While Apple's strategy to offer multiple price options was in play long before the pandemic, one benefit of multiversion pricing is its **resilience to hard times**, said Mr. Srinivasan. "The 'good' option gives firms the opportunity to keep customers within the brand, for those who face economic pressure to downsize their lifestyle," he said.

Meanwhile, Apple's smartphone competitor Samsung Electronics Co. Ltd. launched its flagship Galaxy S20 5G phone in February. As the pandemic months rolled on, the company scrambled to develop a cheaper version, the Samsung Galaxy S20 FE, to appeal to customers on tighter budgets.

Price evaluations

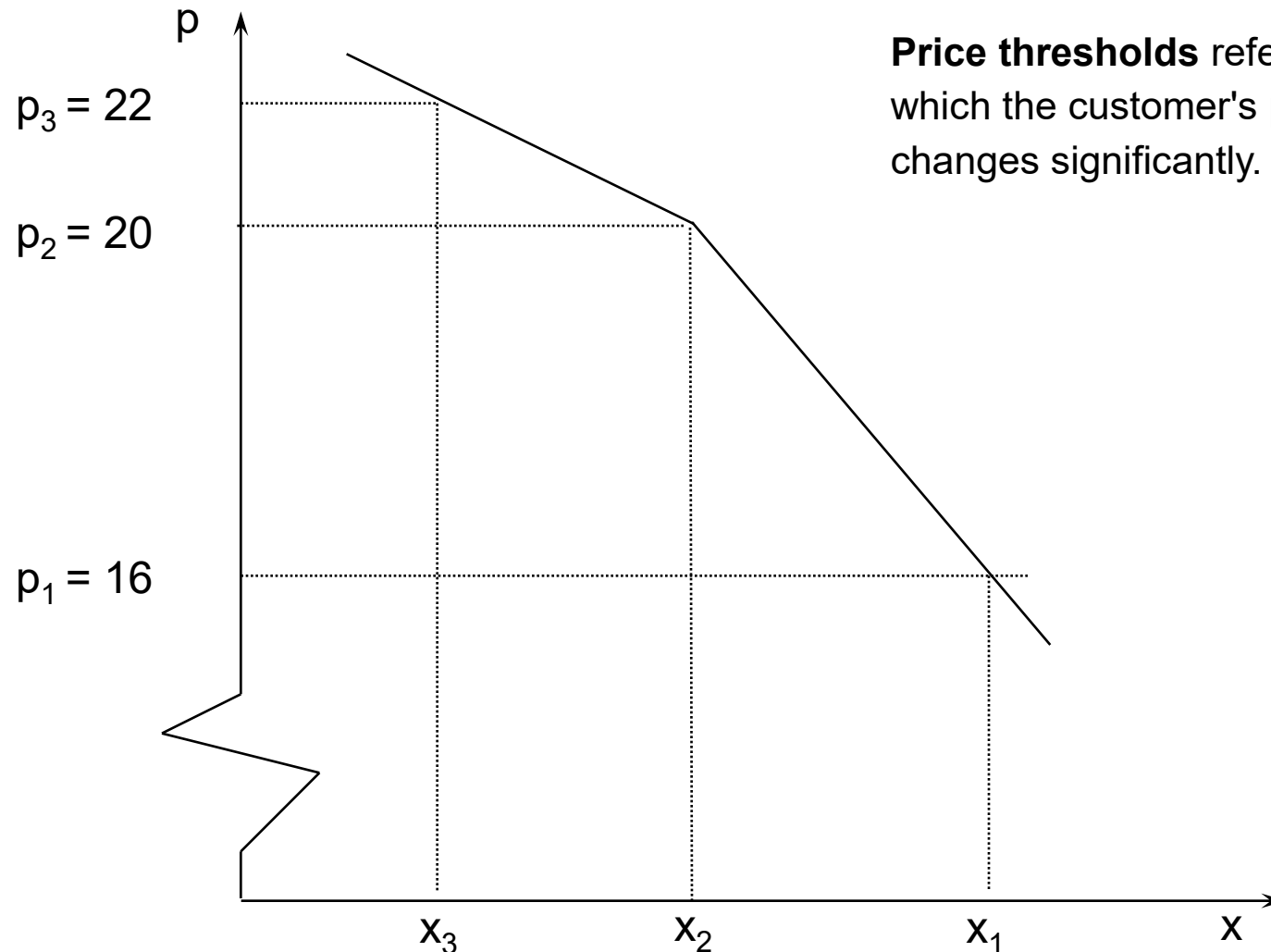
- **Price comparisons:** Customers evaluate products only based on price instead of trading off price against benefits (value for money).
- **Price-dependent quality evaluations:** Customers that are unsure about a product's quality (especially experience goods and credence goods) use product price as an indicator of its quality.
- **Reference prices:** Reference prices refer to the price that is used by the customer as a benchmark for evaluating other prices.

Examples for prices that might influence reference prices

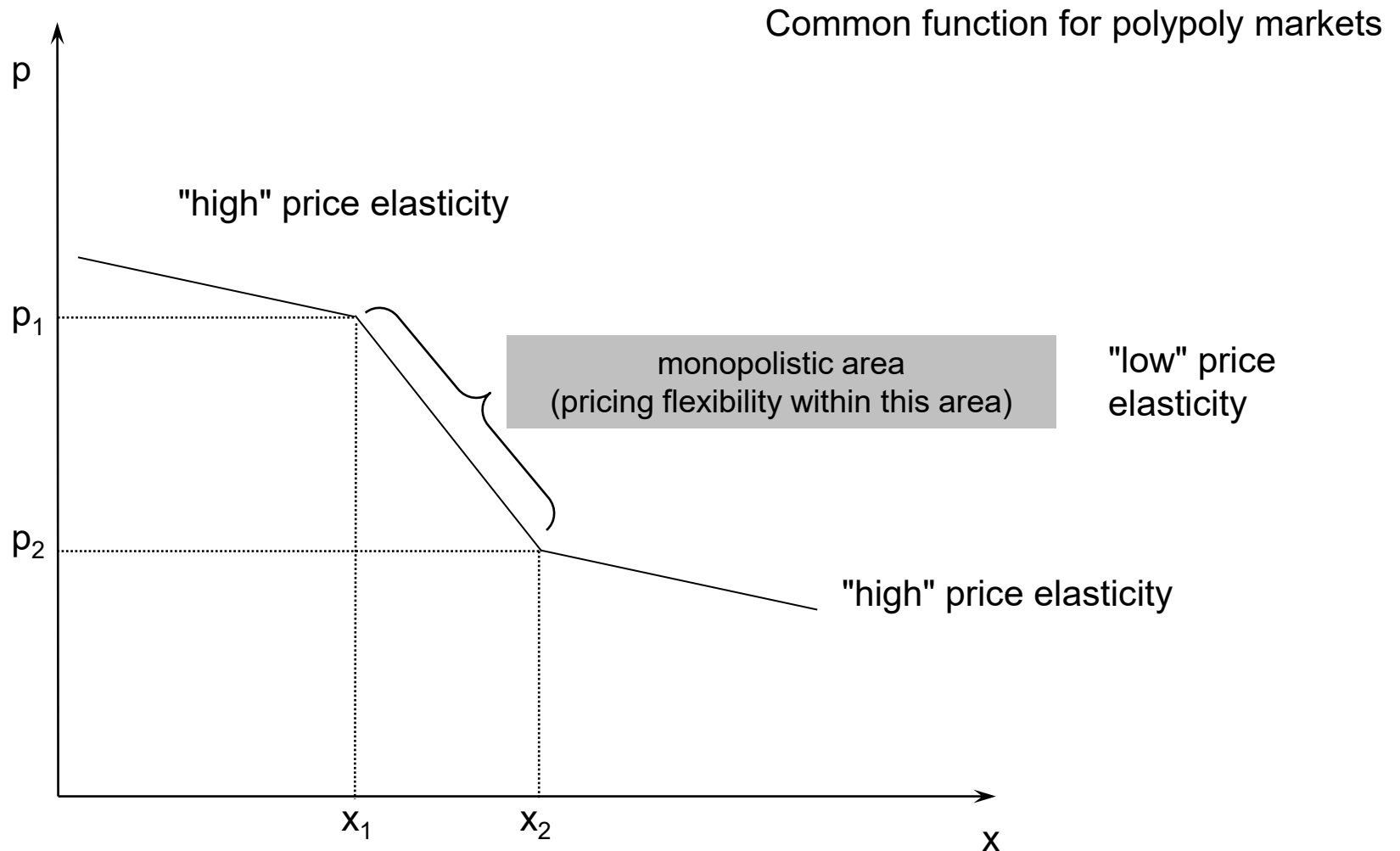
- “Fair price”
- Typical price
- Last price paid
- Competitor prices
- Expected future price
- Usual discounted price

Weber's law: The ability to perceive price differences is limited. Two prices are perceived as different only if a certain threshold is exceeded.

Price evaluations



Price evaluations



Storing of price information

Only some price information is stored in the long-term memory.

Price information is stored for:

- the preferred product
- price information that can be obtained with minimum effort
- prices that are easy to remember (999€, 100€)
- prices that are stable for an extended period of time

→ Result: **price knowledge** (all price-related information (about a product, brand, or retailer) stored in a customer's long-term memory; for instance, reference prices that act as benchmarks when evaluating products)

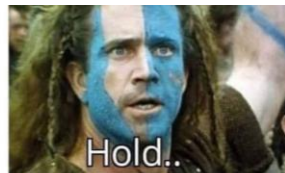
- **Explicit** price knowledge: When a customer is able to state a specific price.
- **Implicit** price knowledge: Prices are remembered vaguely. Customers cannot state specific prices but remember a price evaluation (product A had a 'high' price).

Additional pricing strategies

1. Payment methods (cash vs. credit card/mobile payment)
2. Peanuts effect
3. Communicate absolute or relative price changes
4. Fake pricing / fantasy prices
5. Comparison list price vs. current price

Wednesday

\$ 450,99



Thursday

\$ 450,99



Black Friday

~~\$ 699,99~~

\$ 450,99



Additional pricing strategies

Comparison list price vs. current price



David Ryder | Getty Images

Some deals are too good to be true. And, for **Amazon**, they will cost the company. A Canadian enforcement agency announced today that Amazon **Canada** will pay a \$1 million fine for what could be construed as misleading pricing practices.

The investigation centered on the practice of Amazon **displaying its prices compared to higher "list prices"** — suggested manufacturer prices (MSRPs) **designed as marketing gimmicks** to make people think they are getting a deal, even though it's often the case that **no shopper ever pays that price.**

<http://www.cnbc.com/2017/01/12/amazon-just-got-slapped-with-a-1-million-fine-for-misleading-pricing.html>

Additional pricing strategies

Comparison list price vs. current price ... at amazon.de



Sigma 17-50 mm F2,8 EX DC OS HSM-Objektiv
(77 mm Filtergewinde, für Canon
Objektivbajonett)

Prime Music das erste Mal hören und 3€ Gutscheincode erhalten.
Bedingungen gelten. [Mehr erfahren.](#)

von Sigma

★★★★☆ 352 Kundenrezensionen
| 161 beantwortete Fragen

Amazon's Choice für "sigma 17-50 2.8"

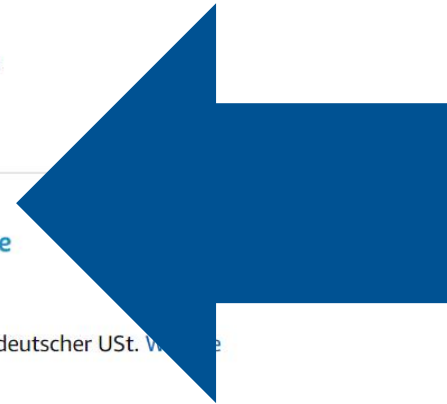
Unverb. Preisempf.: EUR 879,00

Angebot des Tages: **EUR 259,00** ✓prime

Endet in 12h 08m 52s

Sie sparen: **EUR 620,00 (71%)**

Alle Preisangaben inkl. deutscher USt. [Wichtige Informationen.](#)



Additional pricing strategies

Fake pricing / fantasy prices

UK supermarkets criticised over misleading pricing tactics

Which? dossier to competition watchdog detailed 'dodgy multi-buys, shrinking products and baffling offers'



theguardian

... misleading promotional practices such as **“was/now” offers**, where a product is on sale at a discounted price for longer than the higher price applied.

“The CMA’s report confirms what our research over many years has repeatedly highlighted: there are hundreds of misleading offers on the shelves every day that do not comply with the rules. This puts supermarkets on notice to clean up their pricing practices or face legal action.”

<http://www.theguardian.com/business/2015/jul/16/uk-supermarkets-criticised-misleading-pricing>

Price differentiation

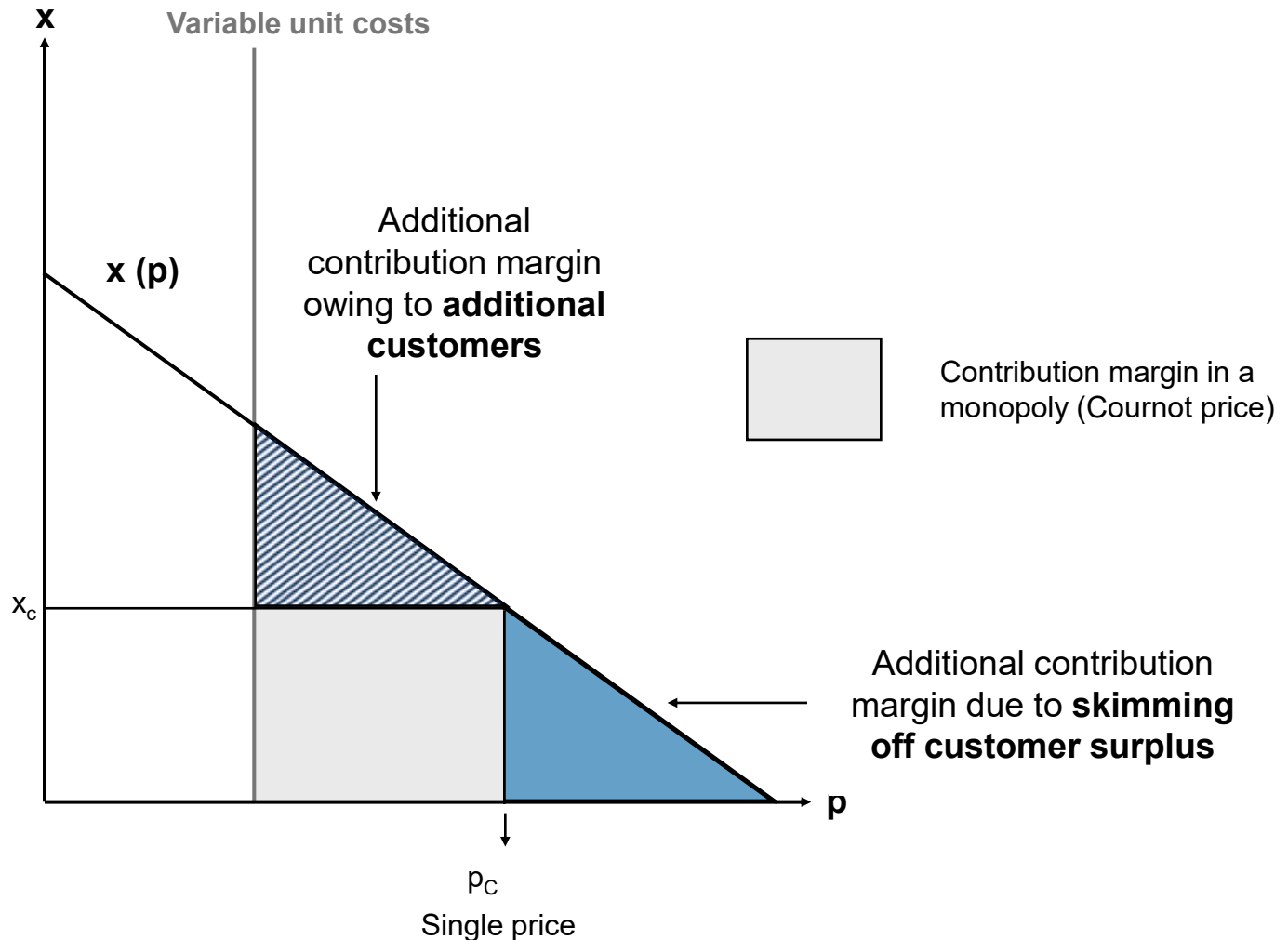
For many markets, customers exhibit a **different willingness to pay for the same product**.
A single price is suboptimal in case of heterogeneity concerning customers' WTP.

→ Solution: Offer the same (or a slightly different) product to different target groups at varying prices.

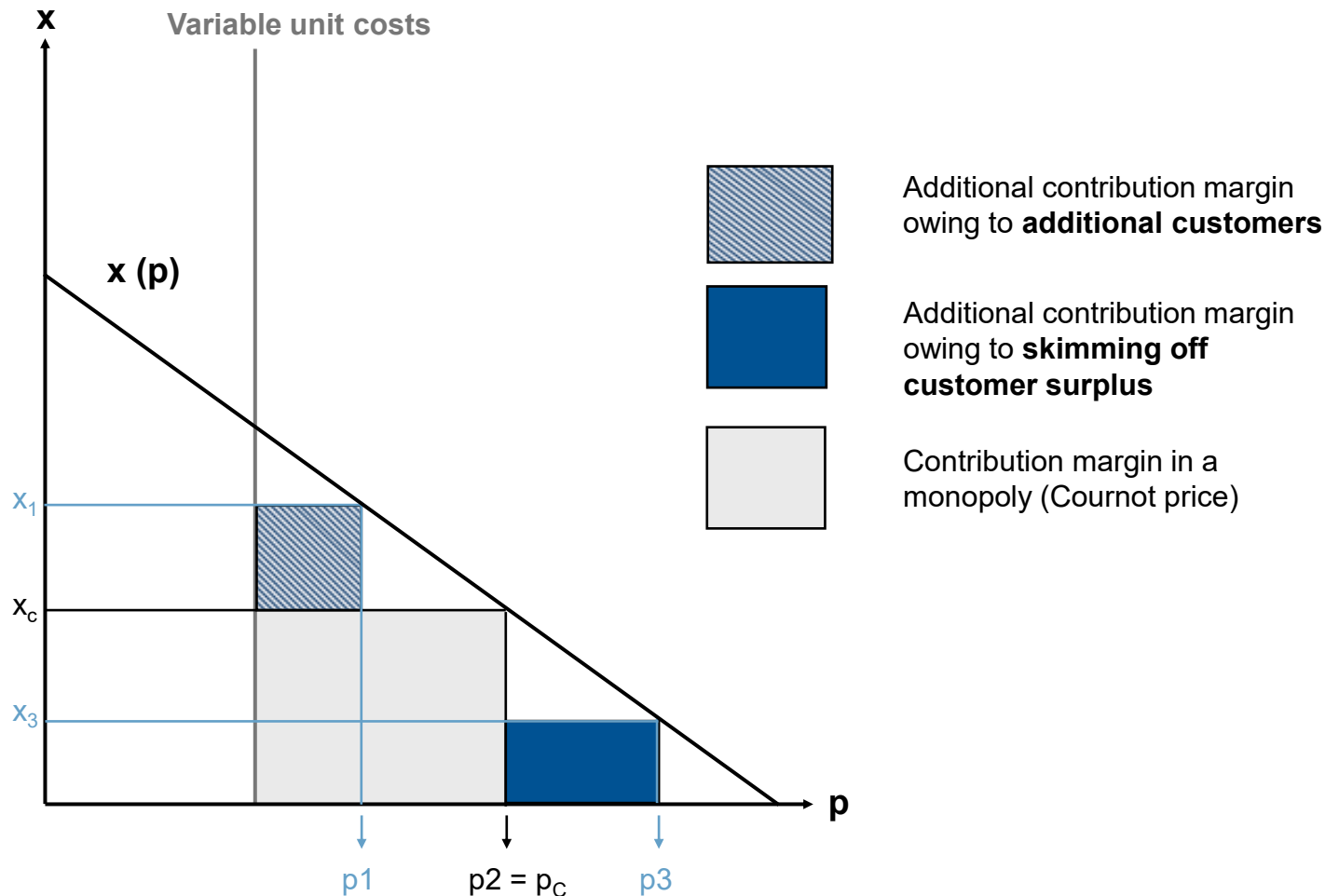
Price differentiation

Context:

- Monopoly
- No fixed costs



Price differentiation



Price differentiation

Preconditions

- Different customer groups exhibit different willingness to pay for the same product
- Customer groups must be identifiable
- Different marketing strategies must be realizable for each target group (e.g., shopping behavior differs significantly)
- Customers cannot easily switch products (e.g., owing to brand loyalty)

Different **types of price differentiation**:

Temporal	airfare, fruits
Geographical	beer (national or regional markets)
Demographic	age (special prices for children), gender (hair salon prices), income (lower prices for students)
Benefit-based	modify add-on benefits such as comfort (economy versus business class)
Quantity-based	volume discounts, loyalty programs
Bundling is a special form of price differentiation	

Temporal price differentiation – Examples



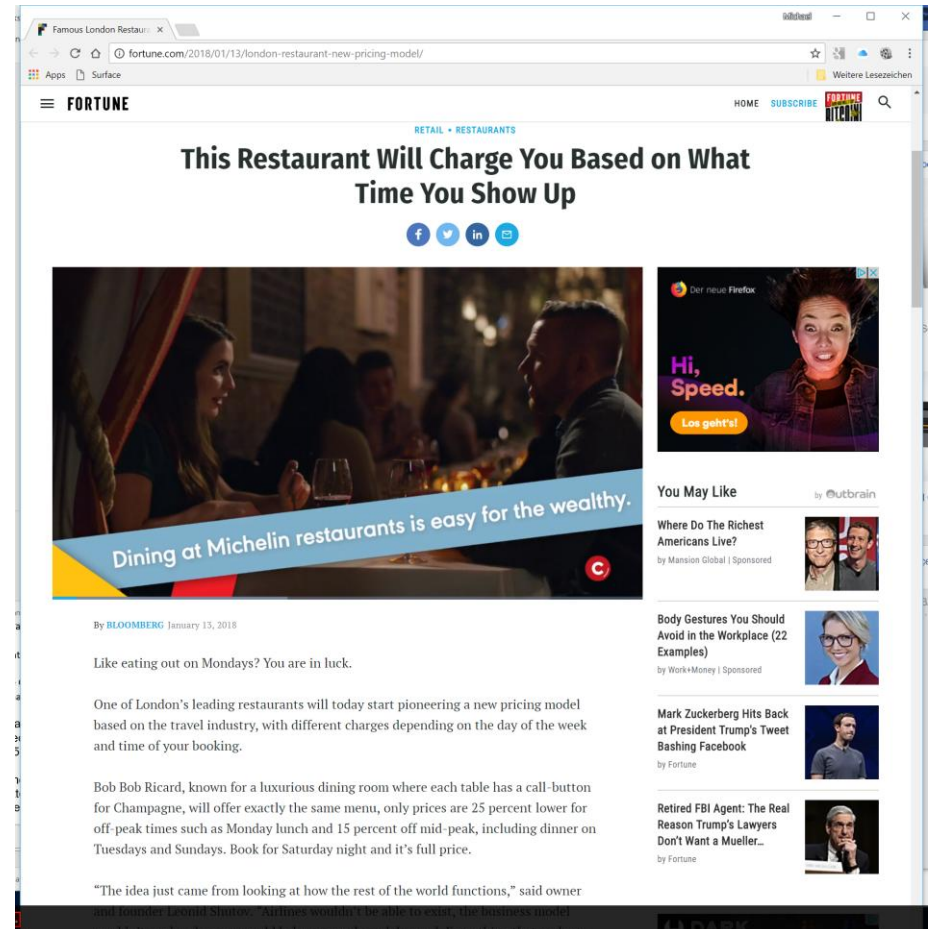
CINEPLEX

Price

Friday, Saturday, Sunday 18.00 €

Monday, Tuesday, Thursday 17.50 €

Happy CINEDAY **Tuesday** 16.00 €



<http://fortune.com/2018/01/13/london-restaurant-new-pricing-model/>

Price differentiation – Dynamic pricing

Surge pricing is a pricing strategy in which companies adjust prices for products or services based on current market demand. It is a model that calculates prices using automatic algorithms.

SURGE PRICING

Demand is off the charts! Rates have increased to get more Ubers on the road.

7.75x

THE NORMAL RATE

\$93 MINIMUM FARE

+

\$5.81 / MIN & \$23.25 / MILE

Sign In
The Washington Post

g+
✉
+ More

Transportation

‘I nearly passed out’: A \$640 Uber ride for a 30-mile trip to the airport

A
🖨
💬 930
🔖 Save for Later
☰ Reading List

U B E R
JANUARY 21, 2016

\$640.94
Ⓢ

09:43am
Dorling, VA

10:32am
Daily, Hevly Parking, Dca, Arlington, VA

CAR	NO. OF	TRIP TIME
SUV	29.80	00:48:52

You accepted 4.4x surge at 9:28am by tapping on this screen. MY FARE WILL BE

4.4

4.4 TIMES THE NORMAL FARE

FARE BREAKDOWN

Base Fare	14.00
Distance	108.77
Time	21.99
Normal Fare	\$144.76
Surge x4.4	492.18
Subtotal	\$636.94
DCA Airport Surcharge (T[email:uber.com])	4.00

LOOKING FOR
THE PERFECT CAR?

Compare top makes and models.

SEARCH NOW

Price differentiation – Dynamic pricing

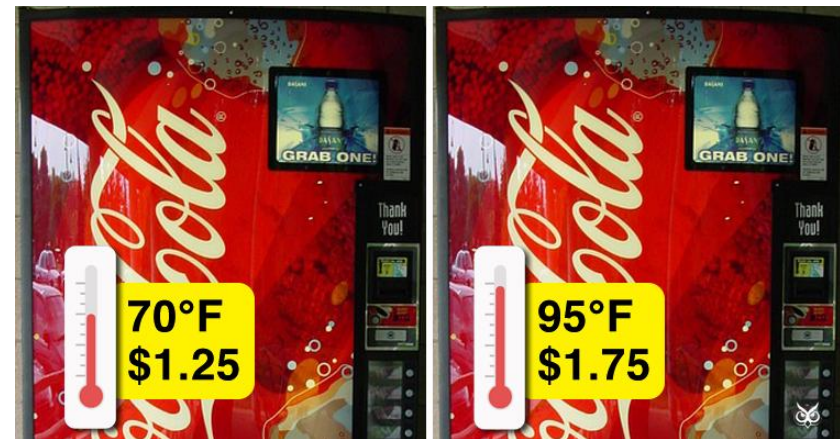
Why Variable Pricing Fails at the Vending Machine (2005)

REMEMBER the plan to charge more for a Coke on a hot day?

A few years back, M. Douglas Ivester, the chief executive of Coca-Cola, said he thought it **would be fair to raise the price of soda on a summer day** like today. Vending machines could be equipped with thermometers, and when demand for a cold soda rose with the temperature, the price would rise too, just as Economics said it should.

It was not one of the great marketing moments in the company's history. In Internet chat rooms and newspaper editorials around the world, angry Coke drinkers denounced the idea. The word "gouging" got tossed around a lot. Pepsi gleefully accused its rival of exploiting consumers.

Coke responded by running away from the heat-seeking vending machine as fast as possible. Company spokesmen said that Mr. Ivester was talking hypothetically and there were no plans to add a summer surcharge. Coke was actually looking for ways that vending machine technology could lower the cost of a drink, they added.



The
New York
Times

<https://www.nytimes.com/2005/06/27/business/why-variable-pricing-fails-at-the-vending-machine.html>

Price differentiation – Yield management

"If I have 2,000 customers on a given route and 400 different prices, I'm obviously short 1,600 [prices]." --- Robert Crandall, former CEO of American Airlines and Wharton MBA.

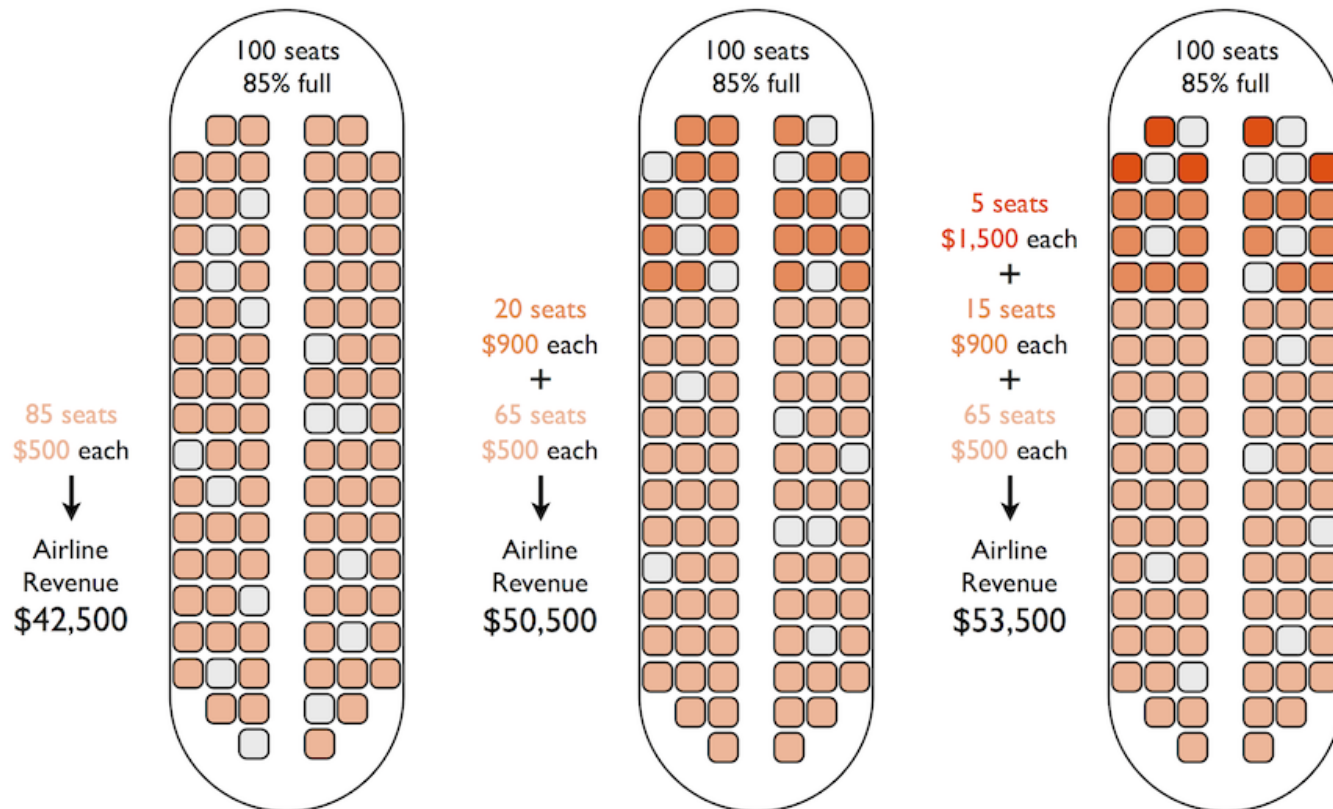
- Yield management is a variable pricing strategy, based on understanding, anticipating and influencing consumer behavior to maximize revenue.
- As an inventory-focused branch of revenue management, yield management involves strategic control of inventory to sell the right product to the right customer at the right time for the right price.

Steps:

- Record demand structure
- Determine customer values
- Forecast of the future demand structure
- Planning of volume/capacity to be provided
- Price determination
- Dynamic adjustment of price and capacity structure based on actual demand or booking

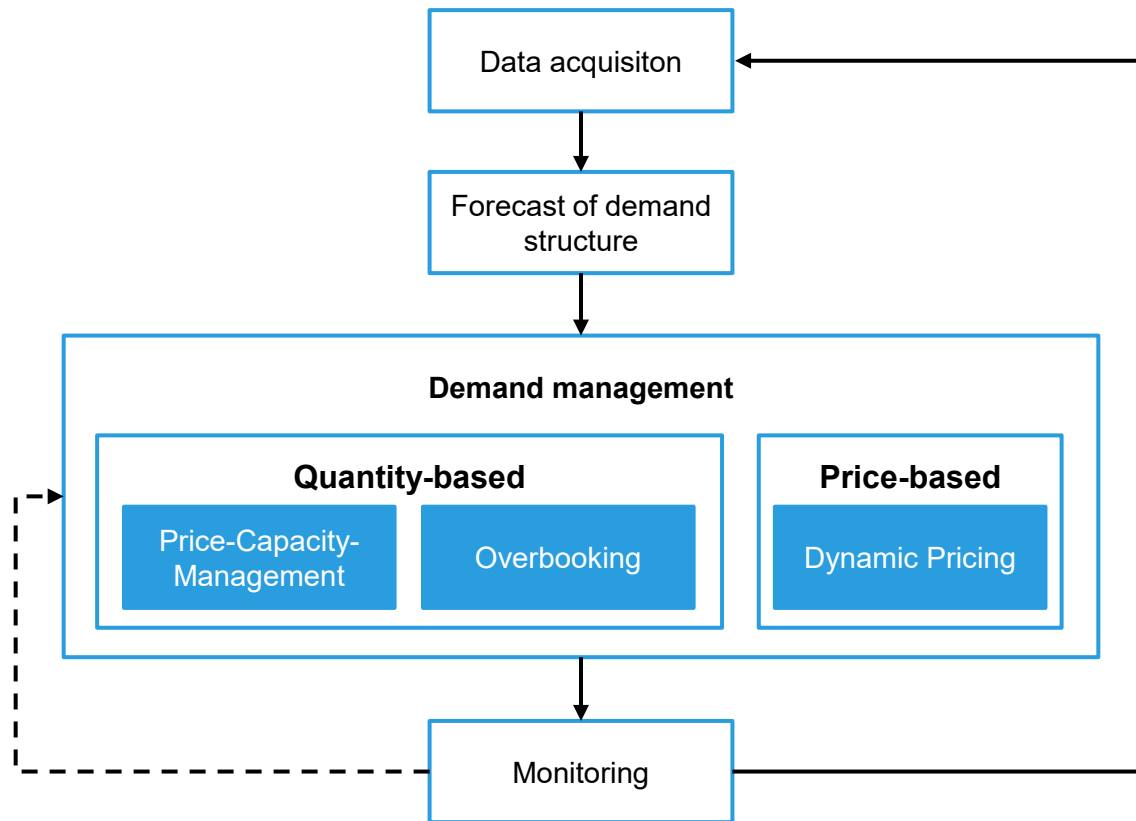
Price differentiation – Yield management

Revenue management process



Price differentiation – Yield management

Revenue management process



Geographical price differentiation – Examples

Different prices depending on the regional market



Geographical price differentiation

— Examples



Geographical price differentiation – Examples

Channel pricing means charging a different price depending on where the consumer buys the product.



0.5 l

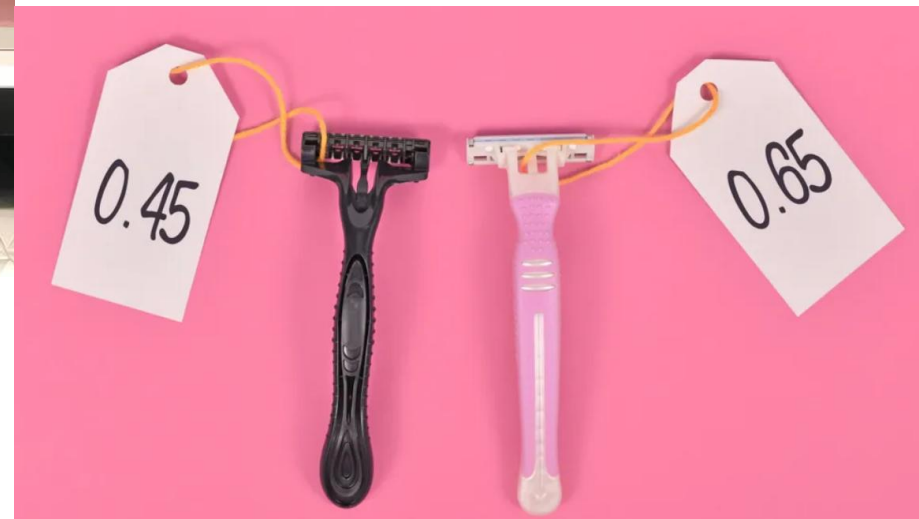
Supermarket 0.79 €

Airport 3.20 €

Demographic price differentiation – Gender



20€ Pink Tax for Prada



Gender Pricing in Germany *

- Fact sheet on the research project -



Studie "Preisdifferenzierung nach Geschlecht in Deutschland" - Forschungsbericht

Iris an der Heiden, Prof. Dr. Maria Wersig;

Die Ergebnisse können Sie im → Steckbrief zum Projekt nachlesen oder die Publikation hier herunterladen.

PDF, File does not meet accessibility standards, 2MB

https://www.antidiskriminierungsstelle.de/SharedDocs/forschungsprojekte/EN/Studie_Preisdifferenzierung_n_Geschlecht_en.html

Gender pricing of products is not significant/very low

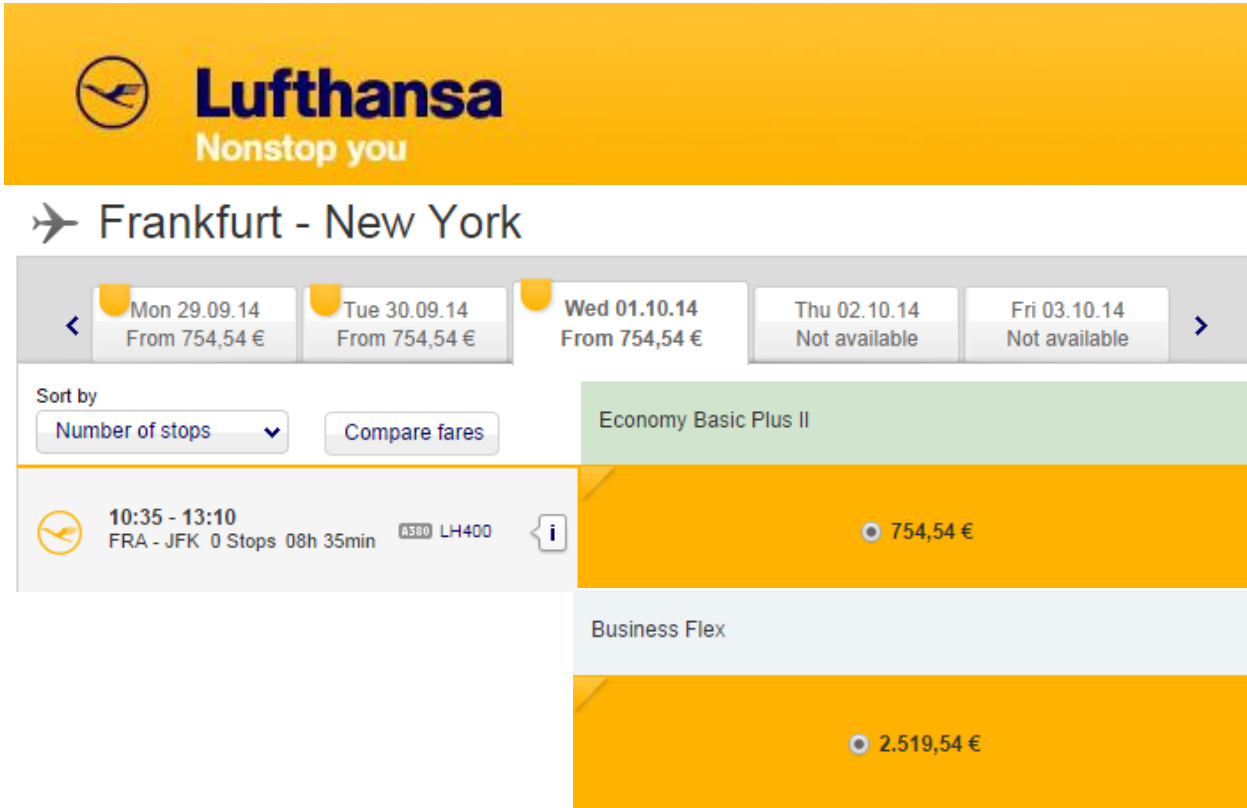
- ▶ The vast majority of products, i.e. altogether 85 per cent, in sectors offering gender-specific product variants, as e.g. children's toys, clothes for adults and children and personal care products, do not have any comparable or similar gender-specific variations.
- ▶ Of the 1,682 products which could be identified as similar in the survey only 3.7 per cent had different prices. In 2.3 per cent of those cases women pay higher prices for the female variant of the same product, in 1.4 per cent of those cases men pay higher prices for the male product variant. The average surcharge for women and men amounts to approximately 5.00 €.

Services increasingly show gender-specific price differences

- ▶ Services as hairdressing, dry cleaning, cosmetic facial treatments and tailoring usually address a specific gender.
- ▶ The survey identified 381 similar services variations for a specific gender, 59 per cent of them revealing a price difference: 50 per cent were more expensive for women, and 9 per cent were more expensive for men.
- ▶ The more obvious gender-based price differences refer to services offered by dry cleaners and hairdressers: 89 per cent of the hairdressers offer different rates for the same short haircuts for women and men, whereas women pay an average surplus amount of 12.50 €. One third of the dry cleaners have different flat rates for men's shirts and women's blouses. The price difference in this case, too, is borne by women, who on average have to pay 1.80 € more for the dry cleaning of blouses than men for shirts.

https://www.antidiskriminierungsstelle.de/SharedDocs/forschungsprojekte/EN/Studie_Preisdifferenzierung_n_Geschlecht_en.html

Benefit-based price differentiation



Lufthansa
Nonstop you

✈ Frankfurt - New York

Mon 29.09.14 From 754,54 € | Tue 30.09.14 From 754,54 € | Wed 01.10.14 From 754,54 € | Thu 02.10.14 Not available | Fri 03.10.14 Not available

Sort by: Number of stops (v) | Compare fares

Flight Details	Class	Price
10:35 - 13:10 FRA - JFK 0 Stops 08h 35min A350 LH400	Economy Basic Plus II	754,54 €
	Business Flex	2.519,54 €

Flight:
Frankfurt to NYC

Economy: €754

Business: €2,519

Price premium:
€1,765

Quantity-based price differentiation



Common quantity-based price differentiation:
Discounts



Quantity-based price differentiation in March 2020:
Corona crisis – quantity-based price **premiums**

Quantity-based price differentiation

Two-part prices

Reason: Decreasing marginal utility of a product

Implementation:

- **Block prices** (e.g., 1-5 products → 10 €/unit, 5-10 products → 8 €/unit, ...)
- **Flatrate** (e.g., „all you can eat“, membership in a fitness club)
- **Two-part prices** (e.g., Bahncard)
- ...



Quantity-based price differentiation

Two-part prices – Restaurant example

- variable costs of a beer are 1 €
- Customers' WTP for a beer are:

1st beer	2nd beer	3rd beer	4th beer	5th beer	6th beer
2.5 €	2 €	1.5 €	1 €	0.5 €	0 €

Optimal solution for an uniform price:

⇒ optimal price: 2 €

⇒ profit: 2 €

Optimal solution for differentiated prices:

⇒ optimal price: 2.5 €, 2 €, 1.5 €, 1 €

⇒ profit: 3 € (+ 50%)

Optimal solution for a two-part pricing system:

⇒ basic fee: 3 €

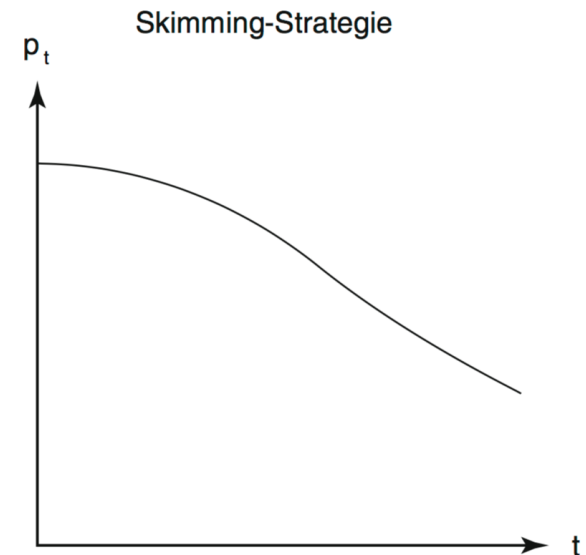
⇒ price for each beer: 1 €

⇒ profit: 3 €

Dynamic pricing strategies

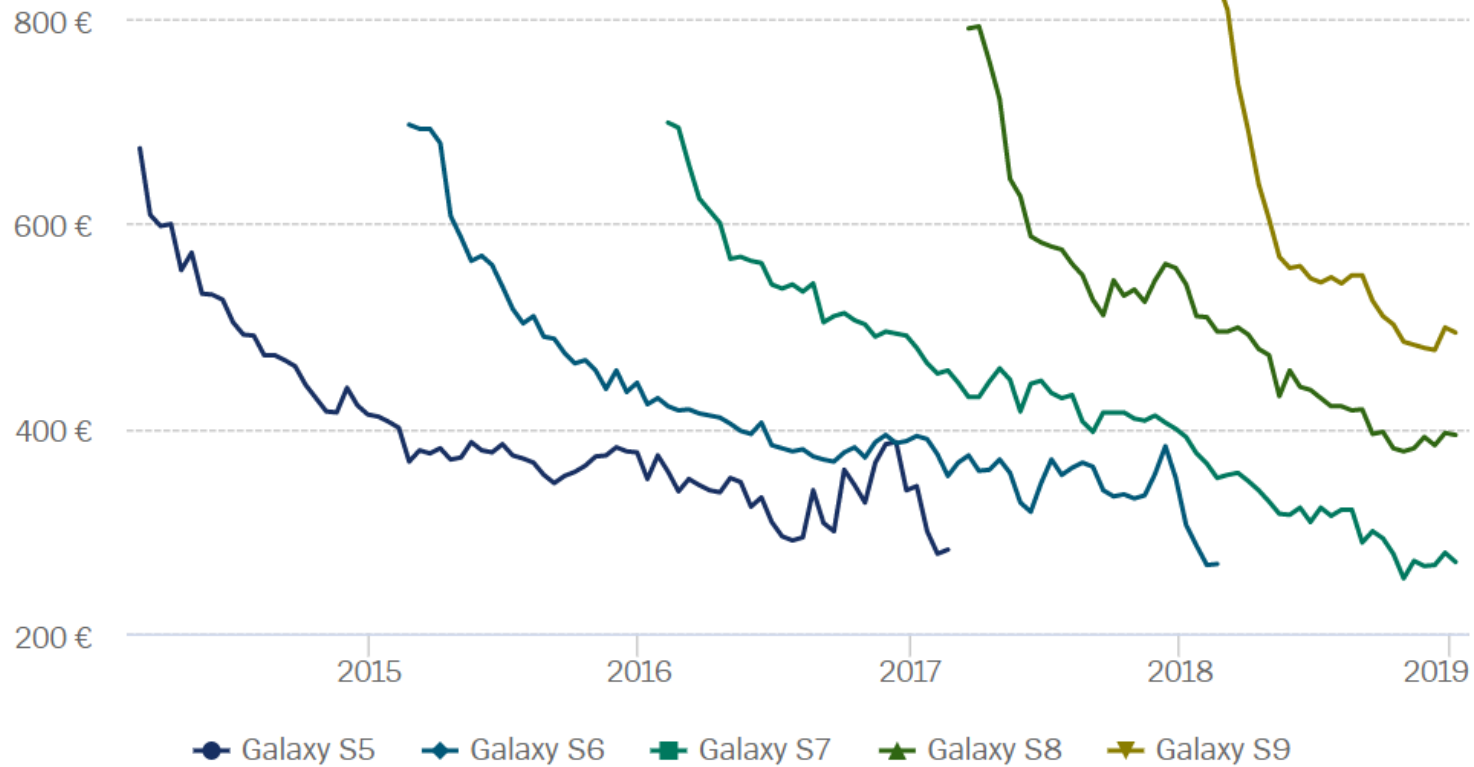
Skimming pricing

- Market entry: Products are sold at a comparably high price when entering a market. Price is reduced over the course of the product lifecycle (owing to competitive pressure)
- take advantage of some buyers' willingness to pay
- investments pay off faster / short product lifecycles
- but... slow innovation diffusion
- establish a positive quality image (owing to a higher price)
- ... you just need the money to finance market growth
(production and distribution capacity is limited)



Dynamic pricing strategies

Skimming pricing – Applied by Samsung

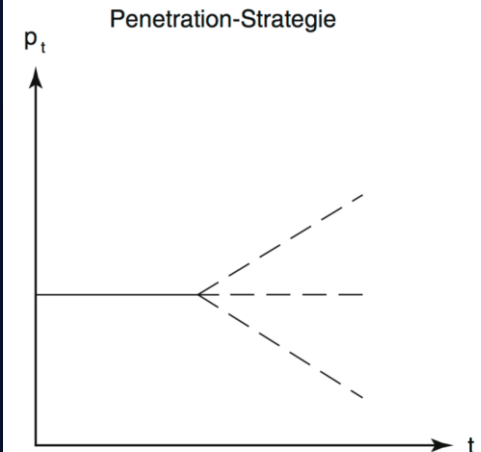


<http://www.spiegel.de/netzwelt/gadgets/samsung-galaxy-s10-wann-sollte-man-es-kaufen-a-1254303.html>

Dynamic pricing strategies

Penetration pricing

- Market entry: establish relatively low prices in order to achieve rapid diffusion of the product and to gain market share
- major goal: establish a strong market position quickly
- profit from economies of scale
- create market entry barriers for competitors
- Price is reduced in case of market entry by competitors



→ Apply this strategy if your customers are very price-sensitive but the market is large enough (mass market)

Price bundling

Principle: Transfer “unused” willingness-to-pay from one product to another
→ Price bundling can lead to cross-selling and increased sales

Example:

- A supermarket offers wine and cheese and has determined the maximum willingness to pay for five consumers.
- The maximum prices reflect the utility of the two products for consumers and result from adding individual willingness to pay (i.e., there is no utility effect resulting from product complementarity)
- Variable unit costs are not taken into account.
- The sales and profit maximum, which is identical in this case, occurs at unit prices:
- p_c (cheese)=5€ and p_w (wine)=4€



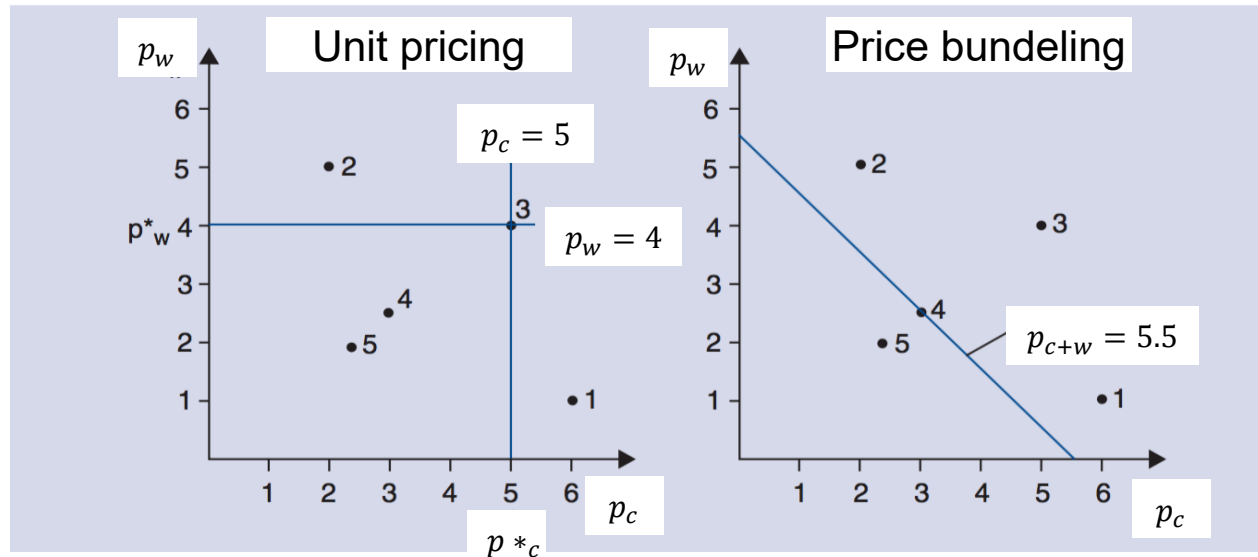
Price bundling

Example: Unit pricing versus price bundling

Consumer	Willingness to pay		
	Cheese	Wine	Bundle
1	6	1	7
2	2	5	7
3	5	4	9
4	3	2,5	5,5
5	2,4	1,8	4,2

Unit pricing: Consumer 3 buys both products, whereas consumers 4 and 5 do not buy any product.

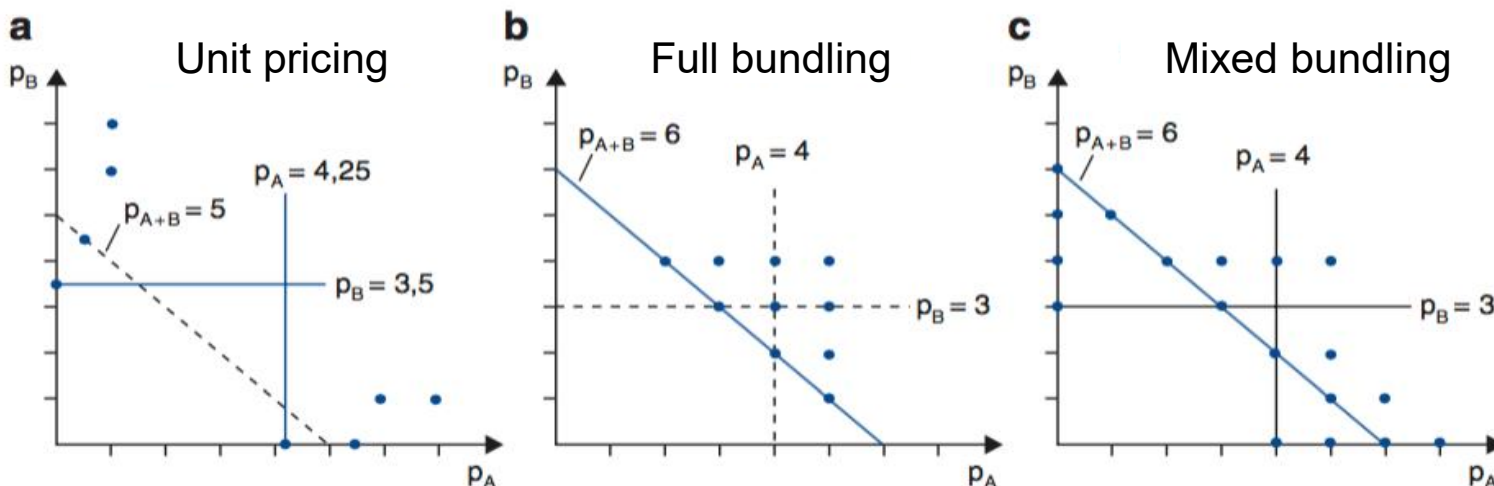
Price bundling: Maximum sales and profits for $p_{c+w} = 5.5$ €. All consumers buy except for consumer 5



Price bundling

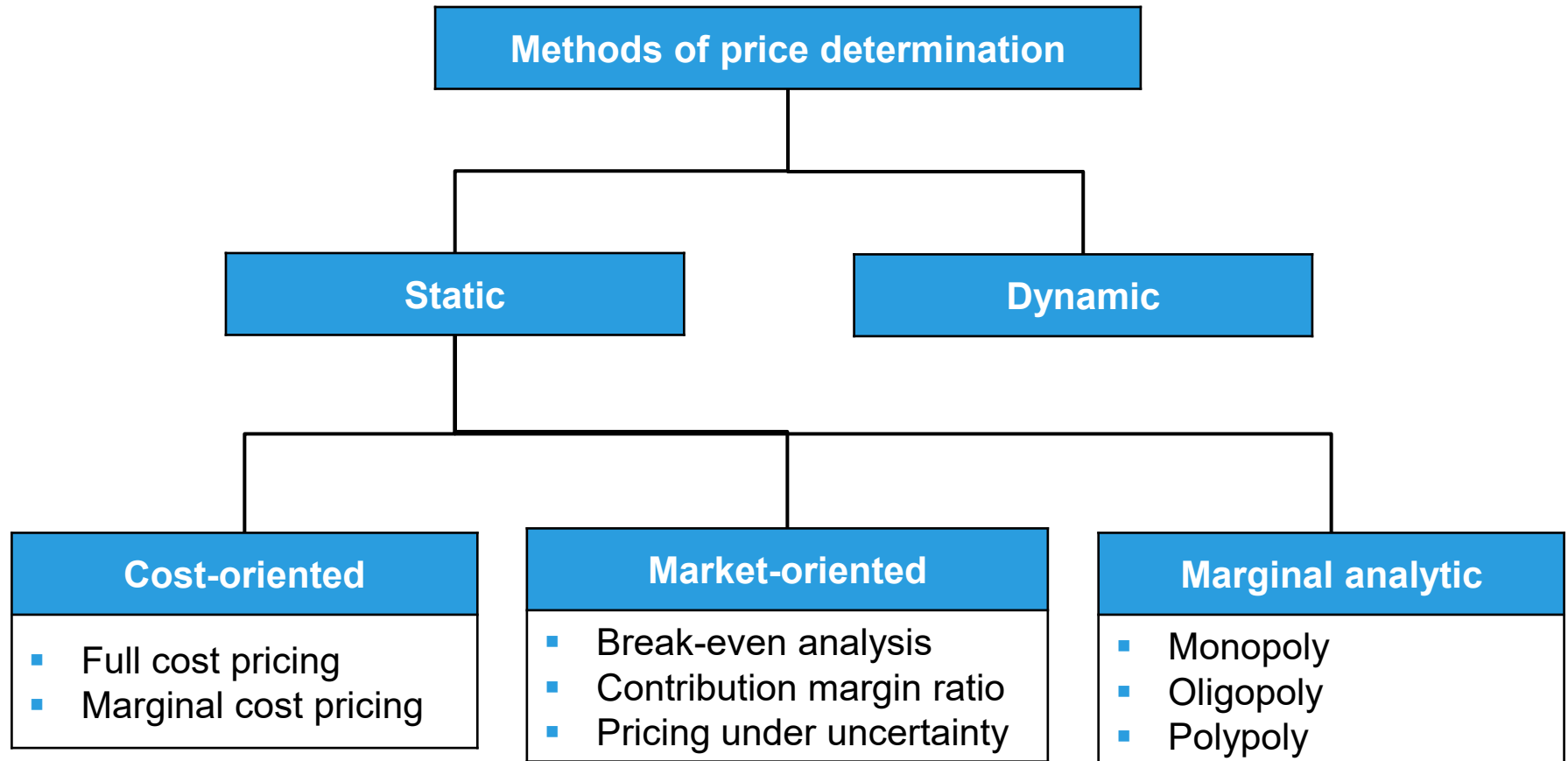
Example: Comparison of unit pricing with full and mixed price bundling

- Unit pricing tends to be advantageous if the utility of the products is very different for consumers: case (a).
- Full price bundling tends to be advantageous if very similar benefits result for both products: Case (b)
- Mixed price bundling tends to be beneficial when case (a) and (b) occur together: Case (c)



• =Willingness to pay of consumer i for products A and B

Methods of price determination



Diller 2008; Olbrich/Battenfeld 2014; Bruhn 2016; Simon/Fassnacht 2016

Full cost pricing

Total unit costs $c_i = C/x$ (with C = total costs, x = number of units sold) are used to determine prices.

Price determination for a product results from the so-called cost-plus pricing

$$p_i = c_i \cdot (1 + \gamma)$$

with

p_i = price for product i

c_i = total unit cost

$\gamma > 0$ = profit margin

Full cost pricing often observed in practice, as all costs are covered by sales and fixed costs are included in the decision.

What are the weaknesses?

Marginal cost pricing

Only the variable unit costs c_{vi} for product i and the required profit margin γ' are used as a basis for price determination.

Cost splitting into fixed and variable costs. Fixed costs are not taken into account.

The price for product i is calculated using the following equation:

$$p_i = c_{vi} \cdot (1 + \gamma')$$

with

p_i = price for product i

c_{vi} = variable unit costs for product i

$\gamma' > 0$ = profit margin

Marginal costing pricing is able to provide well-founded decision support for price determination.

Break-even analysis

Calculates the sales volume required to reach the break-even point at a given price.

Basic assumption: break-even point reached when: cost = sales

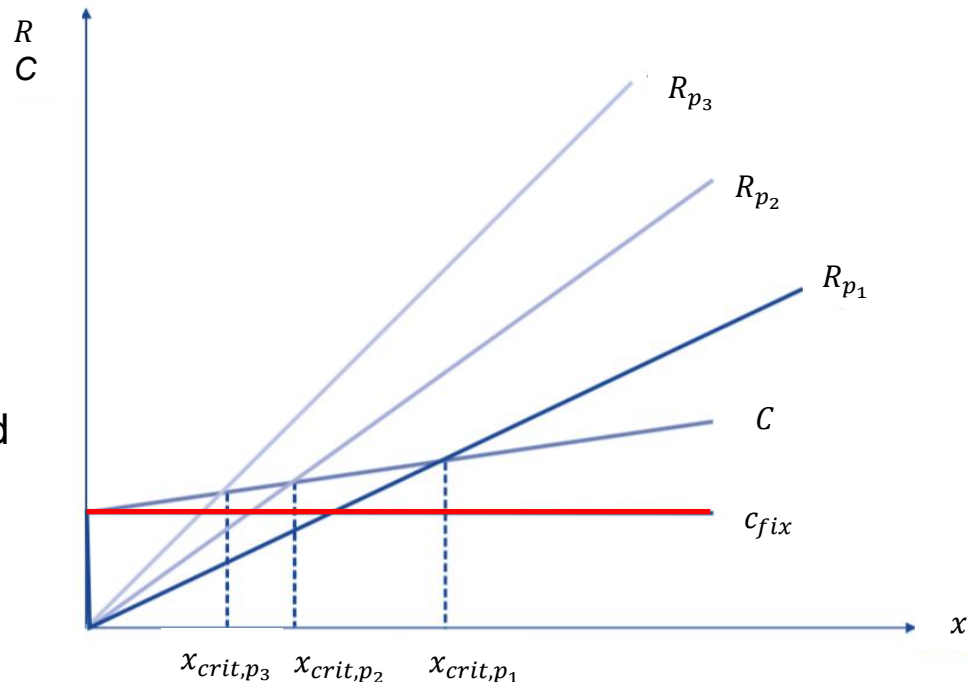
$$C = C_{fix} + c_{var} \cdot x = p \cdot x = R$$

$$x_{crit} = \frac{C_{fix}}{p - c_{var}} \rightarrow \text{Break-even point}$$

x_{crit} = Critical number of units sold

C_{fix} = Fix costs

c_{var} = Variable costs



Contribution margin ratio

Based on break-even analysis

The contribution margin ratio is **the difference between a company's sales and variable expenses, expressed as a percentage**. The total margin generated by an entity represents the total earnings available to pay for fixed expenses and generate a profit.

$$CMR = \frac{p - c_{var}}{p} = \frac{R - C_{var}}{R}$$

The following applies to the break-even point (**R = C**):

$$R_{crit} = \frac{C_{fix}}{CMR}$$

Contribution margin ratio shows the sales volumes, revenues, or prices required to reach break-even.

Market-oriented pricing – An example

Decision situation	S ₋₂	S ₋₁	S ₀	S ₊₁	S ₊₂
Price	16	18	20	22	24
Price change compared to S0	- 20 %	- 10 %	0	+ 10 %	+ 20 %
Variable costs given	12	12	12	12	12
Profit margin	4	6	8	10	12
Contribution margin ratio $CMR = \frac{p - c_{var}}{p}$	0,25	0,33	0,40	0,45	0,50
Fix costs given	400.000	400.000	400.000	400.000	400.000
Profit goal given	150.000	150.000	150.000	150.000	150.000
Revenue needed	2.200.000	1.666.667	1.375.000	1.222.222	1.100.000
Percentage change in sales	+ 60 %	+ 21,2 %	0	- 11,1 %	- 20 %
Sales units needed	137.500	91.667	68.750	55.000	45.833
Percentage change in units	+ 100 %	+ 34,7 %	0	- 19,2 %	- 33,3 %
Profit-turnover ratio	6,8 %	9,0 %	10,9 %	12,3 %	13,6 %

Exercise



The VRG company has developed a new virtual reality headset that will be launched next week. The total fixed cost is 40,000€ and the variable cost is 30€ per unit. The plan is to sell the headset at a price of 80€ per unit.

- Calculate the critical sales volume using break-even analysis. Interpret the result.
- How does the critical sales volume calculated in a) change if the goal is to make a profit of 10,000€ (in addition to just covering the fixed costs)?
- Calculate the contribution margin ratio, derive the break-even point from it and interpret the result.
- What is the relationship between the result calculated in a) and the result calculated in c)?

Pricing under uncertainty

Expected Value Maximization

- Price determination in case of uncertainty about price reactions of competitors
- Risk considerations

$$E_p = \sum_{j=1}^J e_{pj} \cdot w_j \rightarrow \max!$$

with:

E_p = Expected value at price p

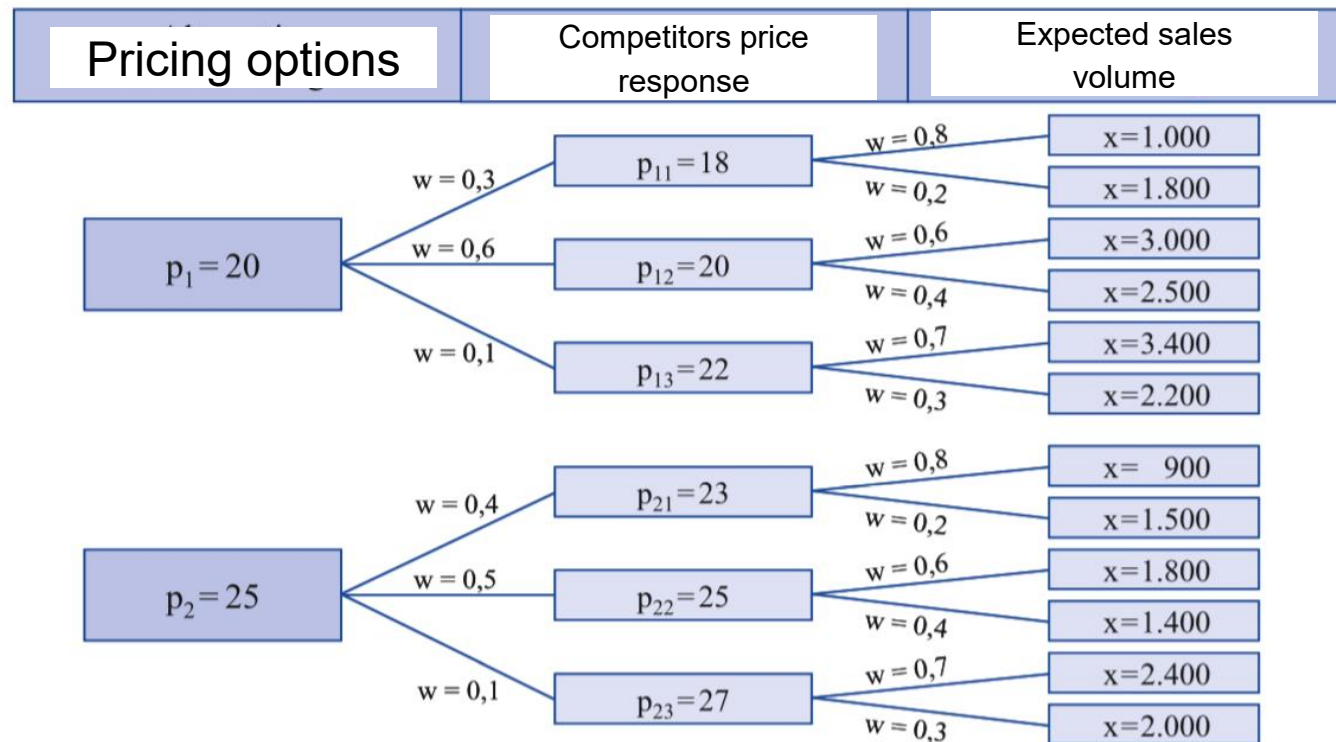
e_{pj} = Outcome at price p in situation j ($j = 1, \dots, J$)

w_j = Probability of situation j (e.g., reaction of competitors)

Possible solution: decision tree

Pricing under uncertainty

Decision tree for pricing decisions



$$E(p_1) = (1.000 \cdot 0,8 + 1.800 \cdot 0,2) \cdot 0,3 + (3.000 \cdot 0,6 + 2.500 \cdot 0,4) \cdot 0,6 + (3.400 \cdot 0,7 + 2.200 \cdot 0,3) \cdot 0,1 = 2.332$$

$$E(p_2) = (900 \cdot 0,8 + 1.500 \cdot 0,2) \cdot 0,4 + (1.800 \cdot 0,6 + 1.400 \cdot 0,4) \cdot 0,5 + (2.400 \cdot 0,7 + 2.000 \cdot 0,3) \cdot 0,1 = 1.456$$

Exercise

A publisher wants to set the price for a new book. A competitor is expected to publish a book on a very similar topic soon, but the pricing of the competitor's book is unclear. Thus, the sales volume of the publisher's book depends on the own and the competitor's price.

The publisher thinks that three different prices p_i are reasonable $p_1 = 19\text{€}$, $p_2 = 22\text{€}$ or $p_3 = 25\text{€}$. The own price will affect the competitor's pricing p_{ij} .

With the help of an experienced marketing manager, possible competitor price responses and their probabilities are determined. For the different price constellations of the two companies, the marketing manager also makes forecasts for the expected sales volumes and their probabilities.

The marketing manager estimates, for example, that if the company sets the own price to 19€, the competitor will set the price of his book to 18€ with a probability of 20%. In this price constellation, 1000 units of the book will be sold with a probability of 80%.

All estimates provided by the marketing manager are shown on the following slides.

Exercise

Table 1: Probabilities of competitor price p_{ij} at own price p_i

	$p_{i1}=18$	$p_{i2}=19$	$p_{i3}=20$	$p_{i4}=22$	$p_{i5}=25$
$p_1=19$	0,2	0,8	-	-	-
$p_2=22$	-	-	0,3	0,4	0,3
$p_3=25$	-	-	-	-	1

Exercise

Table 2: Probabilities for expected sales volumes at own price p_i and competitor price p_{ij}

Sales volume (probability) at p_i and p_{ij}	$p_{i1}=18$	$p_{i2}=19$	$p_{i3}=20$	$p_{i4}=22$	$p_{i5}=25$
$p_1=19$	1.000 (0,8) 1.500 (0,2)	1.300 (0,6) 1.600 (0,4)	1.500 (0,6) 1.600 (0,4)	1.500 (0,4) 1.900 (0,6)	2.500 (0,6) 1.900 (0,4)
$p_2=22$	500 (0,8) 300 (0,2)	700 (0,7) 900 (0,3)	600 (0,7) 800 (0,3)	1.200 (0,8) 1.500 (0,2)	2.400 (0,7) 2.000 (0,3)
$p_3=25$	400 (0,6) 200 (0,4)	500 (0,9) 300 (0,1)	500 (0,9) 900 (0,1)	800 (0,6) 900 (0,4)	1.000 (0,6) 900 (0,4)

Exercise

Your tasks:

- a. Draw a decision tree showing the relationship between the publisher's own alternative prices and the expected sales quantities!
- b. Which own price leads to the highest possible sales volume if the method of expected value maximization is applied?
- c. Show that your own price of 25€ leads to the highest possible profit if variable costs are 16€.

Market structures

Different market structures can be observed in practice:

Monopoly: competitors are either non-existent or their impact is negligible; companies thus only need to consider customer reactions

Oligopoly: many small customers and a small number of sellers (oligopolists); companies must consider both customer reactions and competitors' actions

Polypoly: many small customers and many small suppliers; companies have no flexibility in pricing; a competition-based pricing approach may be necessary

Monopoly: seller can define supply (number of units produced) and define prices to maximize profits.

→ **Price function**= price p depends on the desired demand x

Marginal analytic price determination

Price-sales function (PAF)

- (Functional) relationship between expected sales volume and price
- Market reaction to price changes
- Basic assumption: Sales volume decreases with price increase

Linear PAF

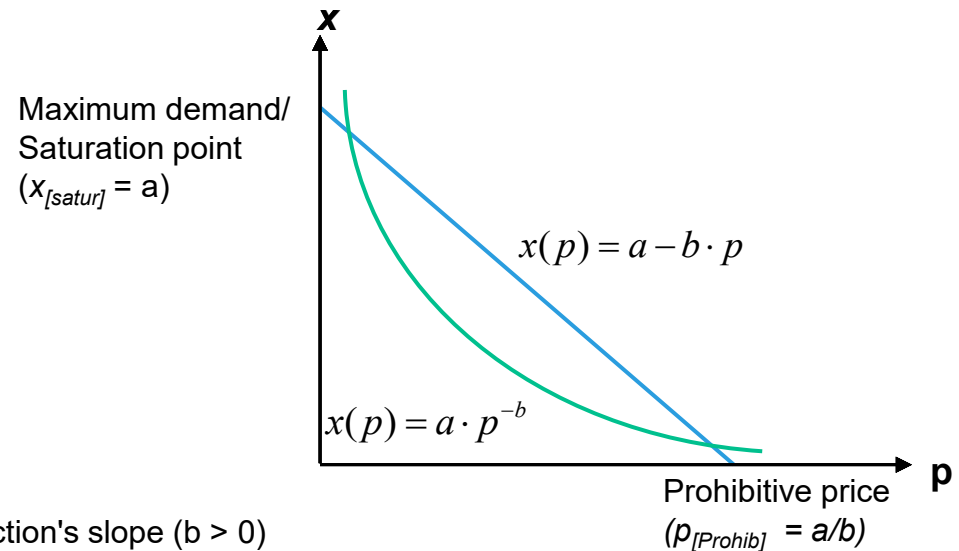
$$x(p) = a - b \cdot p$$

$$p(x) = a - b \cdot x$$

Multiplicative PAF (Cobb-Douglas)

$$x(p) = a \cdot p^{-b}$$

with p = Price a = Constant
 x = Number of units sold b = price function's slope ($b > 0$)



Marginal analytic price determination

Price elasticity of demand

Consumers' reaction (sales volume) to changes in price.

"By what percentage does sales volume x change if price p changes by 1%?"

Direct price elasticity

$$\epsilon_{x(p),p} = \frac{\frac{\Delta x(p)}{x(p)}}{\frac{\Delta p}{p}} = \frac{\Delta x(p)}{\Delta p} \cdot \frac{p}{x(p)}$$

Inelastic demand: $|\epsilon_{x(p),p}| < 1$
 elastic demand: $|\epsilon_{x(p),p}| > 1$

Price point elasticity

$$\epsilon_{x(p),p} = \lim_{\Delta p \rightarrow 0} \frac{\frac{\Delta x(p)}{x(p)}}{\frac{\Delta p}{p}} = \frac{dx(p)}{dp} \cdot \frac{p}{x(p)}$$

Gives an indication how a particular price change will affect sales

Price point elasticity for linear PAF (Monopoly):

$$\epsilon_{x(p),p} = \frac{-bp}{a - bp}$$

Decker et al. 2015; Bruhn 2016

Marginal analytic price determination

Static-optimal price (monopoly)

Approach to determine optimal price: $Profit(p) = R(p) - C(x(p)) = p \cdot x(p) - C(x(p)) \rightarrow \max!$

→ **Marginal revenue = Marginal cost**

$$\frac{dProfit(p)}{dp} = \overbrace{x(p) + p \cdot \frac{dx(p)}{dp}}^{\text{Marginal revenue}} - \overbrace{\frac{dC(x(p))}{dx(p)} \cdot \frac{dx(p)}{dp}}^{\text{Marginal cost}} = 0$$

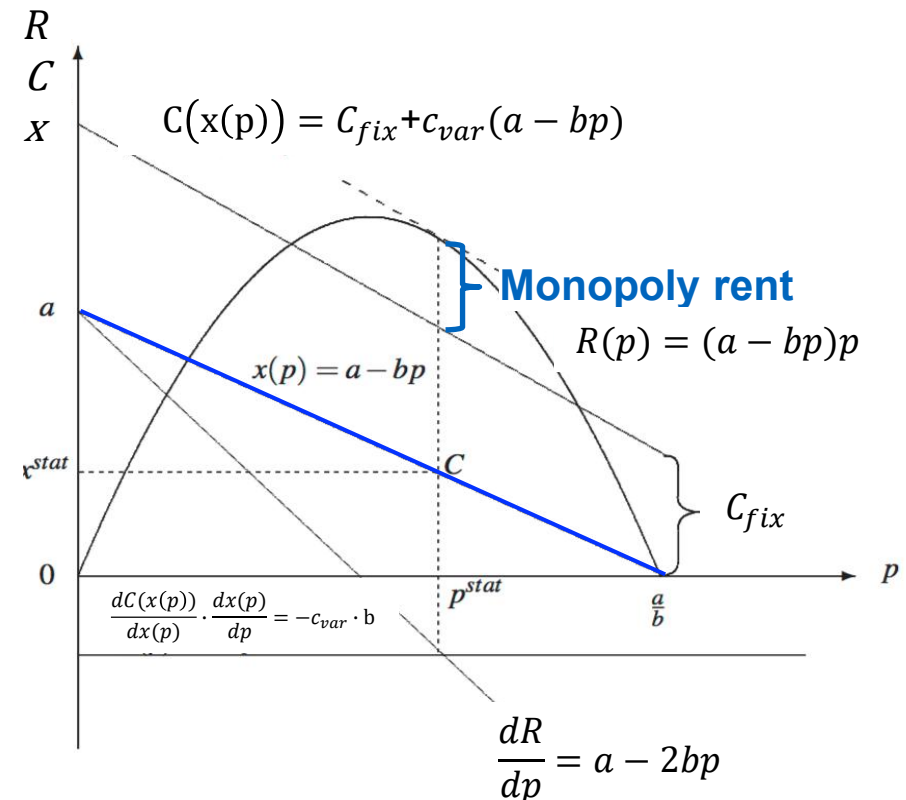
$$p^{stat} = \frac{\varepsilon}{1 + \varepsilon} C' \quad C' = \frac{dC(x(p))}{dx} = c_{var}$$

With **linear PAF** and **linear cost function**:

$$Profit(p) = (a - bp)p - (C_{fix} + c_{var}(a - bp))$$

$$p^{stat} = \frac{1}{2} \left(\frac{a}{b} + c_{var} \right) \quad \text{Cournot price}$$

$$x^{stat} = \frac{1}{2} (a - b \cdot c_{var}) \quad \text{Cournot quantity}$$



Exercise

The company PapaTürk has launched a lemonade that combats bad breath after eating garlic. Given the patenting of the strictly held recipe, the company acts as a monopolist to its potential customers with this product.

The variable costs for a filled can (=1 unit) are 80 cents. The fixed costs for the next planning period is 3,000 €. The following PAF indicates the relationship between price and sales quantity:

$$p = 5 - \frac{x}{3000}$$



Exercise

- a. What is the prohibitive price in this market and is its meaning?
- b. What is saturation point assumed with the present function for the new brand?
- c. Which price leads to the maximum revenue?
- d. The company wants to maximize its profit. Calculate the profit-maximizing quantity and the corresponding profit-maximizing price!
- e. Calculate the profit that results from maximizing revenue and profits according to the results of tasks c) and d)!



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